ASYS 245 · AIRCRAFT SYSTEMS 1

Exam 1 Review

Modules 1–6 · Atmosphere → High-Speed Flight

Name _____ Partner _____

Your 75 minutes

0–5
Rate yourself. Start with your weakest module.

5–50
Work 2–3 modules from memory first (pencil). Stuck? Leave it, keep going. Then check the site and fix in a different colour.

50–65
Partner challenges + quiz each other on the vocab.

65–75
Re-rate. Circle 3 things to review tonight.

Module	Confidence before (1–5)	Confidence after (1–5)	Mock Qs right
1 · Atmosphere			/ 13
2 · Lift & Drag			/ 12
3 · Stability & Control			/ 10
4 · Flight Controls			/ 14
5 · Phases of Flight			/ 14
6 · Performance: Stall & High-Speed Flight			/ 14

Why this works

- **Pulling it from memory** beats re-reading the notes. Being tested on material produces far better long-term recall than studying it again (Roediger & Karpicke, 2006).
- A major review of study methods rated **quizzing yourself** as one of the two most effective strategies — and **re-reading and highlighting** as low value (Dunlosky et al., 2013).
- A wrong answer you fix today is one you get right on the exam.

Full Cornell notes & flashcards

Three full mock exams (30 Qs each)

Built from the ASYS 245 study site (Cornell notes + Mock Exams A, B, C). These are the essentials only — the full notes are on the site. Weight & Balance (Module 7) is not on Exam 1.

FIX IT My error log

Every miss goes here — this page is your study list for tonight

Got one wrong or left it blank? Don't just fix the letter. Write down **why** the right answer is right — that's the part that sticks.

Module · Q#	What I put / what I thought	The right answer	Why it's right (in my own words)

3 things to review tonight

1. _____
2. _____
3. _____

Questions for my instructor

1. _____
2. _____
3. _____

MODULE 1 Atmosphere

Full notes, flashcards & mock exams on the site

Big ideas — read, then fold this box back and work from memory

- Dry air = **78% nitrogen, 21% oxygen, 1% other** (mostly argon).
- A gas is described by 4 things: **pressure, temperature, mass, volume**. Change one and another changes.
- **Temp ↑ → volume ↑ → density ↓ → performance ↓** (less lift, less thrust, less engine power). Cold air = better performance.
- **Altitude ↑ → less air above → pressure ↓ → gas expands.** Air pressure = the weight of the air above you.
- **Bernoulli:** fluid speeds up → static pressure drops. Fast air over the curved top = low pressure on top = lift.

Vocabulary — write the letter of the matching term

- ☐ Volume is proportional to temperature when pressure is held constant (V = cT).
- ☐ One goes up, the other goes down.
- ☐ Faster-moving fluid → lower pressure. Total = dynamic + static.
- ☐ Pressure and volume are inverse when temperature is held constant (p × V = C).
- ☐ Both rise or fall together by the same ratio.
- ☐ Mass ÷ volume. Cold air has more of it than warm air.
- ☐ The sea-level reference everyone compares weather and performance to.
- ☐ The weight of the air above a location; drops as you climb.

A Boyle's Law
B Charles' Law
C Density
D Air pressure
E ISA / standard day
F Bernoulli's principle
G Inversely
H Proportional

Gas laws

Law	Formula	What is held constant	Relationship
Boyle's Law	_____	_____	_____ — V ↑ means p ____
Charles' Law	_____	_____	_____ — T ↑ means V ____
Density	D = _____	—	Cold air = _____ density = _____ performance

Standard day (ISA) at sea level — know all three units

Item	Value
Pressure	_____ psi = _____ inHg = _____ mb
Temperature	____ °C = ____ °F
Every 1,000 ft you climb (rule of thumb)	About ____ °C colder and about ____ inHg less pressure
Units spelled out	psi = _____ · inHg = _____ · mb = _____

Arrow chains — fill in ↑ or ↓

Start	Then	Then	Result
Temperature ↑	Volume ____	Density ____	Aircraft performance ____
Altitude ↑	Air above you ____	Air pressure ____	Volume of a sealed balloon ____
Air speed over wing top ↑	Dynamic pressure ____	Static pressure ____	Lift ____

Quick check

1. Boyle's Law: $p_1 = 3$, $V_1 = 8$. Pressure rises to $p_2 = 6$. V_2 = ____ ($p_1 \times V_1 = p_2 \times V_2$)
2. Medicine Hat airport is 2,350 ft. Standard pressure there ≈ 29.92 − _____ ≈ _____ inHg.
3. Bernoulli: Total pressure = _____ pressure + _____ pressure. The airspeed indicator reads _____ pressure.

Mock exam questions — circle one answer

1. Apart from nitrogen and oxygen, the remaining 1% of dry air is mostly: Mock A

a Argon **b** Water vapour **c** Hydrogen **d** Carbon monoxide

2. A sealed balloon is carried from sea level up to 10,000 ft and its temperature does not change. Its volume will: Mock A

a Increase **b** Drop to zero **c** Stay the same **d** Decrease

3. On a standard day, temperature falls by about how much for every 1,000 ft climbed? Mock A

a 1 °C **b** 2 °C **c** 5 °C **d** 10 °C

4. The airspeed indicator works by measuring: Mock A

a Outside air temperature at the pitot tube
b Air density in the pitot line
c Static pressure only, from the static port
d Dynamic pressure (total minus static)

5. Air pressure at any location is best described as: Mock B

a The density of the air at ground level
b The speed of the wind
c The weight of the column of air above it
d The temperature of the air above it

6. Boyle's Law holds which property constant? Mock B

a Density **b** Temperature **c** Pressure **d** Volume

7. A pressure of 29.92 inches of mercury is the same as: Mock B

a 14.7 psi **b** 59 mb **c** 1013.2 psi **d** 15 inHg

8. Compared with a cold morning, on a hot afternoon the same aircraft will have: Mock B

a A shorter take-off run because the engine is warm
b Poorer performance because the air is less dense
c Exactly the same performance, since weight is unchanged
d Better climb performance because warm air rises

9. Air flows through a venturi tube. At the narrowest point the air's: Mock B

a Speed and static pressure both increase
b Speed decreases and static pressure increases
c Speed increases and static pressure decreases
d Temperature rises sharply

10. Charles' Law says that when a gas is heated at constant pressure, its volume: Mock C

a Increases **b** Decreases **c** Stays the same **d** Drops to zero

11. The standard sea-level temperature of 15 °C is equal to: Mock C

a 32 °F **b** 52 °F **c** 59 °F **d** 68 °F

12. Air pressure is lower at 10,000 ft than at sea level because: Mock C

a The air is warmer higher up, so it expands
b The percentage of oxygen is lower at altitude
c Gravity stops pulling on air above 10,000 ft
d There is less air above, so less weight pressing down

13. Medicine Hat airport is about 2,350 ft above sea level. On a standard day its air pressure is about: Mock C

a 14.7 inHg **b** 27.5 inHg **c** 29.92 inHg **d** 31.3 inHg

Partner challenge (2 min): Explain to your partner why a Cessna 172 climbs better on a cold January morning than a hot July afternoon. You must use the words **density** and **volume**.

MODULE 2 Lift & Drag

Full notes, flashcards & mock exams on the site

Big ideas — read, then fold this box back and work from memory

- 4 forces: **Lift** (90° to flight path, through the CP), **Weight** (down, through the CG), **Thrust** (forward), **Drag** (back). Level flight: L = W and T = D.
- Lift two ways, both true: **Newton** — the wing turns air down, air pushes the wing up. **Bernoulli** — faster air on top = lower pressure on top.
- Lift equation: **double the speed → 4× the lift** (speed is squared). Double the area → 2×. Half the density → half.
- **AOA ↑ → lift ↑ until the critical angle (≈ 15–18°)** → airflow separates → stall.
- **Total drag = parasite (↑ with speed) + induced (↓ with speed)** → a U-shaped curve. Least drag where they cross.

Vocabulary — write the letter of the matching term

- ☐ Sudden loss of lift when the critical angle of attack is exceeded.
- ☐ The one point lift acts through; it moves as angle of attack changes.
- ☐ Extra lift and less induced drag within about one wingspan of the ground.
- ☐ The point weight acts through; shifts as fuel burns.
- ☐ Span² ÷ area. High = long and narrow = less induced drag.
- ☐ Thin layer of air stuck to the surface — laminar at the front, turbulent farther back.
- ☐ Where the boundary layer changes from laminar to turbulent.
- ☐ Point where the pitching moment stays the same, about ¼ chord back.

A Centre of pressure (CP)
B Centre of gravity (CG)
C Aerodynamic centre
D Stall
E Boundary layer
F Transition point
G Ground effect
H Aspect ratio

Label the airfoil

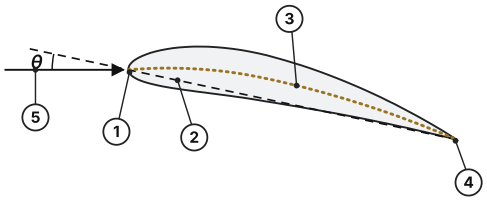


Figure · Airfoil cross-section

Word bank: Leading edge · Trailing edge · Chord line · Mean camber line · Relative wind · Angle of attack

1 _____
2 _____
3 _____
4 _____
5 _____
θ _____

If the airfoil were symmetrical, line 3 would lie right on top of line ____ (no camber).

Drag — 5 names, 2 families

Drag type	Family (parasite or induced)	What causes it
Skin friction	_____	_____
Form (pressure)	_____	_____
Interference	_____	_____
Induced	_____	_____
As airspeed ↑ …	Parasite ____ Induced ____	Total drag curve is ____-shaped

The three axes — all pass through the CG

Axis	Runs from …	Motion	Control	Stability it gives
Lateral	_____	_____	_____	_____
Longitudinal	_____	_____	_____	_____
Vertical	_____	_____	_____	_____

Wing & tail design

Design	What it is	What it does
Dihedral	Wings angled ____	_____ roll stability — the low wing makes more lift
Anhedral	Wings angled _____	Less roll stability, more _____ (fighters)
Winglet	Small fin at the wingtip	Blocks the vortex → less _____ drag
Swept wing	Wing angled back	Cuts drag near the _____
T-tail	Stabilizer on _____	Out of the wing's wake, but _____

Quick check

1. Airspeed doubles, nothing else changes → lift is ____ times as great. Wing area doubles → lift is ____ times as great.
2. Steady, level flight: Lift = _____ and Thrust = _____.

Mock exam questions — circle one answer

1. The point on the chord where the pitching moment stays the same as angle of attack changes is the: Mock A

a Centre of gravity **b** Transition point
c Centre of pressure **d** Aerodynamic centre

2. An aircraft's airspeed doubles and nothing else changes. Lift becomes: Mock A

a Half as great **b** Twice as great **c** Four times as great
d Unchanged

3. Flying within about one wingspan of the ground causes: Mock A

a Less induced drag and more lift
b More induced drag and less lift **c** A shock stall
d A higher stall speed

4. A long, narrow wing (high aspect ratio) mainly reduces: Mock A

a Interference drag **b** Skin friction drag **c** Induced drag
d Form drag

5. Wings that angle downward from the root to the tip are called: Mock A

a Sweepback **b** Flat wing **c** Anhedral **d** Dihedral

6. As airspeed increases in level flight, induced drag: Mock B

a Increases **b** Stays the same **c** Becomes zero
d Decreases

7. Compared with a conventional tail, a T-tail: Mock B

a Sits out of the wing's wake, but needs a stronger, heavier fin
b Is lighter and simpler, because the fin carries less load
c Provides lateral stability instead of pitch stability
d Cannot have an elevator, so the whole tail must move

8. Using Newton's laws, a wing makes lift by: Mock C

a Turning (deflecting) the air downward, so the air pushes the wing up
b Creating a vacuum above the wing that sucks it upward
c Using its own weight to press down on the air below
d Making the air above the wing denser than the air below

9. The critical angle of attack for most light-aircraft wings is about: Mock C

a 4–6° **b** 15–18° **c** 30–35° **d** 45°

10. Interference drag is caused by: Mock C

a Air rubbing on a rough or dirty skin surface
b Shock waves forming on the wing near Mach 1
c Airflows from two parts meeting (e.g. wing–fuselage joint)
d Producing lift, through the wingtip vortices

11. The point where the boundary layer changes from laminar to turbulent is the: Mock C

a Aerodynamic centre **b** Stagnation point
c Centre of pressure **d** Transition point

12. Fast aircraft use swept wings mainly to: Mock C

a Make the wing lighter and easier to build
b Increase interference drag at the wing root
c Reduce drag as the aircraft nears the speed of sound
d Increase lift at low speed for take-off

Partner challenge (2 min): Stand up and be the aircraft: your partner calls out **pitch, roll, yaw** — you name the axis, the control, and the stability. Swap after 6 calls.

MODULE 3 Stability & Control

Full notes, flashcards & mock exams on the site

Big ideas — read, then fold this box back and work from memory

- **Stability** = what the aircraft does on its own after a disturbance. **Control** = the pilot making it move.
- Name trap: **lateral stability = roll** (longitudinal axis). **Longitudinal stability = pitch** (lateral axis). **Directional = yaw.**
- Pitch: the wing's lift is behind the CG → nose wants to drop → the **tail pushes down** to balance, like a seesaw.
- Roll: **dihedral** — the low wing meets the air at a higher AOA → more lift → rolls level. Yaw: the **fin weathercocks** the nose back.
- **Adverse yaw:** the rising wing has more lift AND more induced drag → the nose yaws opposite the roll. Fix: rudder into the roll.

Vocabulary — write the letter of the matching term

- ☐ Slow, diverging spiral dive when directional stability beats lateral.
- ☐ Rolling-and-yawing wallow when dihedral effect beats directional stability.
- ☐ Fin angled slightly off centre to line up with the spiralling propeller slipstream.
- ☐ Upward wing angle; the low wing makes more lift and rolls the aircraft level.
- ☐ Airflow hits the side of the fin and swings the nose back into the relative wind.
- ☐ Rudder used with aileron so roll and yaw go together and the ball stays centred.
- ☐ Stalled-wing wake blankets a T-tail so the elevator cannot lower the nose.
- ☐ Small horizontal surface ahead of the main wing.

A Dihedral
B Weathercock effect
C Offset fin
D Dutch roll
E Spiral instability
F Deep stall
G Coordinated turn
H Canard

Read the traces — pitch attitude after a gust

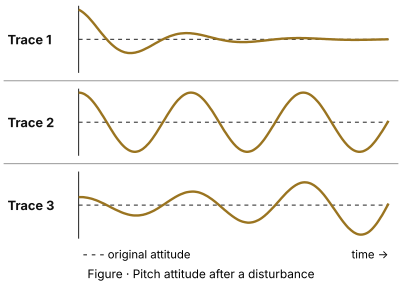


Figure · Pitch attitude after a disturbance

Word bank: Positive · Neutral · Negative

Trace 1 _____ dynamic stability

Trace 2 _____ dynamic stability

Trace 3 _____ dynamic stability

Static vs dynamic stability

	Positive	Neutral	Negative
Static — the first tendency			
Dynamic — over time			

Three kinds of stability

Stability	Motion	Works around which axis	Provided by
Longitudinal			+ CG in range
Lateral			
Directional			

Four design fixes for adverse yaw — name each one

How it works	Name of the fix
The UP aileron travels farther than the DOWN aileron	_____
The up-going aileron's nose pokes below the wing into the airflow	_____
Springs move the rudder automatically with the ailerons	_____
A panel rises on the down-going wing to kill its lift (large jets)	_____

Compare

	Cause	Fix / result
Dutch roll	_____ beats _____	Fixed by a _____
Spiral instability	_____ beats _____	Slow, diverging _____
CG too far AFT	—	Aircraft is _____
CG too far FORWARD	—	Nose _____, _____ stall speed

Quick check

1. Roll RIGHT with ailerons only → left aileron goes _____ → left wing gets more lift and more _____ → nose yaws _____.

Mock exam questions — circle one answer

1. After a gust, an aircraft simply holds its new attitude — it neither returns nor keeps moving away. Its static stability is: Mock A

a Dynamic **b** Negative **c** Positive **d** Neutral

2. Dutch roll happens when: Mock A

a Lateral stability (dihedral effect) is stronger than directional stability
b The CG is too far forward
c Directional stability is stronger than lateral stability
d The wings have anhedral

3. A CG located too far aft will make the aircraft: Mock A

a Longitudinally unstable **b** Unable to roll
c Nose heavy with a higher stall speed
d More stable but sluggish in pitch

4. Which adverse-yaw fix makes the up-going aileron travel farther than the down-going aileron? Mock A

a Roll spoilers **b** Differential ailerons
c Aileron–rudder interconnect **d** Frise ailerons

5. On a conventional aircraft in cruise, the horizontal tail normally produces: Mock B

a No force at all, because it is set at zero angle
b Forward thrust from the air flowing off the wing
c A downward force that balances the wing's nose-down tendency
d An upward lifting force that helps carry the weight

6. After a gust, an aircraft starts back toward level flight, but every swing past level is bigger than the one before. It has: Mock B

a Positive static and positive dynamic stability
b Positive static and negative dynamic stability
c Negative static and neutral dynamic stability
d Neutral static stability only

7. A pilot rolls to the LEFT using ailerons only. Adverse yaw will swing the nose: Mock B

a Right **b** Up **c** Left **d** Down

8. The "weathercock" effect describes: Mock C

a The fin pushing the tail so the nose swings back into the relative wind
b Adverse yaw pulling the nose away from the turn
c The wings rolling level by themselves after a gust
d The nose dropping on its own when the wing stalls

9. A CG located too far forward will cause: Mock C

a Spiral instability in a steep turn
b Instability in pitch and a flat-spin risk
c A higher stall speed and a hard time raising the nose
d A lower stall speed and a lighter elevator

10. A lifting canard differs from a conventional tail because it: Mock C

a Pushes downward to hold the nose up, like a tail
b Is mounted on top of the fin, out of the wing wake
c Controls roll instead of pitch
d Produces an up-load, sharing the weight with the main wing

Partner challenge (2 min): A rolls a hand-aircraft left, no rudder. B says which way the nose yaws and why, then names two fixes.

MODULE 4 Flight Controls

Full notes, flashcards & mock exams on the site

Big ideas — read, then fold this box back and work from memory

- **Primary controls:** ailerons → roll · elevator → pitch · rudder → yaw. All three work by **changing the camber** of the surface.
- **Secondary controls:** flaps, slats, slots, spoilers, speed brakes, trim. They change lift, drag or control force.
- **Trim tab moves OPPOSITE** to where you want the surface to go. Trim removes pilot effort — it doesn't add control.
- **Mass balance** = weight ahead of the hinge → stops **flutter**. After a repair or repaint, the surface must be re-balanced.
- **Flaps ↑ camber → more lift + more drag.** First ~20° = mostly lift (take-off); beyond = mostly drag (landing).

Vocabulary — write the letter of the matching term

☐ Weight ahead of the hinge line so the surface's CG is on or in front of the hinge.
☐ Drag-only panel, placed so it adds no pitch, roll or yaw.
☐ Fixed gap in the leading edge that lets high-pressure air over the top to delay the stall.
☐ Part of the surface ahead of the hinge so the airflow helps move it — lighter controls.
☐ Movable leading-edge piece that slides forward to open a slot only when needed.
☐ Self-feeding vibration of a control surface — can tear it off.
☐ Panels that pop up on touchdown to dump lift so the brakes work.
☐ One-piece horizontal tail that pivots as a whole.

A Stabilator
B Flutter
C Mass balance
D Horn balance
E Slot
F Slat
G Ground spoiler
H Speed brake

Label the airliner

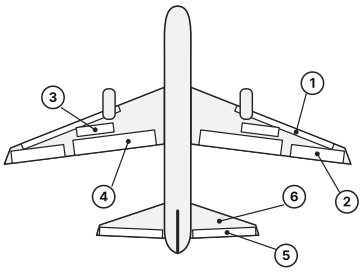


Figure · Airliner seen from above

Word bank: Aileron · Elevator · Horizontal stabilizer · Trailing edge flap · Leading edge slat · Flight spoiler

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____

Which number is the fixed surface that gives longitudinal (pitch) stability? ____

Primary flight controls

Control	Axis	Motion	Pilot input → result
Ailerons	_____	_____	Down aileron = _____ lift → that wing _____
Elevator	_____	_____	Stick back → elevator ____ → nose ____
Rudder	_____	_____	Right pedal → rudder right → nose _____

Tabs — which way does it move compared with the surface?

Tab	Same or opposite?	Its job
Trim tab	_____	_____
Anti-servo tab (on a stabilator)	_____	_____
Servo / balance tab	_____	_____
Fixed tab	Bent on the ground by _____	_____

Flaps — simplest to most effective

#	Flap type	How it works
1	_____	The whole trailing edge hinges down. Cheapest, least effective.
2	_____	Only the _____ surface hinges down. More drag than lift.
3	_____	A gap lets high-pressure air flow over the flap → keeps flow attached.
4	_____	Slides back and down on tracks → adds _____ and camber. Most lift.

Quick check

1. Elevator trim tab moves UP → airflow pushes the elevator _____ → nose _____ trim.
2. Flight spoilers raised on ONE wing help the _____ roll. Raised on BOTH wings → descend without _____.

Mock exam questions — circle one answer

1. A weight fitted ahead of a control surface's hinge line is there to: Mock A

a Prevent flutter **b** Trim the aircraft for hands-off flight
c Reduce adverse yaw **d** Increase the surface's camber

2. A fixed trim tab on an aileron is adjusted by: Mock A

a Turning the trim wheel in the cockpit
b An AME bending it on the ground, then test-flying
c An electric motor and a switch
d The autopilot during cruise

3. Ground spoilers deploy on touchdown mainly to: Mock B

a Dump lift so the weight goes onto the wheels and the brakes work
b Reduce adverse yaw during the landing roll-out
c Increase lift for a softer, smoother touchdown
d Trim the aircraft nose-up for the landing roll

4. A horn balance on a rudder or elevator is there to: Mock B

a Add weight behind the hinge line
b Use the airflow ahead of the hinge to make the control lighter
c Make the surface stall at a higher angle
d Hold the surface in place for hands-off flight

5. Which flap slides back and down on tracks, increasing both camber and wing area? Mock C

a Plain flap **b** Split flap **c** Slotted flap **d** Fowler flap

6. Which leading-edge device is movable and opens a slot only when it is needed? Mock C

a Slat **b** Fixed slot **c** Spoiler **d** Fowler flap

7. Compared with a spoiler, a speed brake: Mock C

a Is fitted only to gliders
b Adds drag only and is placed so it causes no pitch, roll or yaw
c Raises on one wing at a time
d Deploys automatically on touchdown

8. A split flap differs from a plain flap because: Mock C

a It slides aft on tracks and adds wing area
b A slot opens between the wing and the flap
c Only the lower surface of the trailing edge hinges down
d It is fitted to the leading edge

9. A stabilator is: Mock C

a A one-piece horizontal tail that pivots as a whole
b A fixed stabilizer with a hinged elevator
c A leading-edge device that delays the stall
d A balance weight ahead of the hinge

10. Match each description to the correct device. Each letter is used once; two letters are not used. Mock B

A Trim tab · **B** Servo tab · **C** Anti-servo tab · **D** Fixed tab · **E** Mass balance · **F** Spoiler · **G** Horn balance

- ☐ Moves the SAME way as a stabilator to add control feel
- ☐ Moves opposite the control surface to hold it in place for hands-off flight; set from the cockpit
- ☐ Deflects opposite the surface so the airflow pushes a heavy surface for the pilot (big aircraft, no boost)
- ☐ Small metal tab bent on the ground to cure a wing-heavy tendency
- ☐ Weight ahead of the hinge line that stops flutter

Partner challenge (2 min): AME check: your partner just repainted an elevator. In 30 seconds, tell them what they must do before it flies, and why. (Hint: flutter.)

MODULE 5 Phases of Flight

Full notes, flashcards & mock exams on the site

Big ideas — read, then fold this box back and work from memory

- Three phases: **climb** (with take-off), **cruise**, **descent** (with landing).
- Bottom of the total-drag U = **L/D max** — parasite = induced. Best speed for **longest glide AND maximum range**.
- **Vx** = most height per **distance** (clear the trees). **Vy** = most height per **time** (normal climb). Vx is slower.
- **Power available vs power required:** extra = climb · equal = level · short = descend. The gap sets the rate.
- Engine running → set airspeed with **power**. Engine failed → set airspeed with **attitude** (nose down = faster).

Vocabulary — write the letter of the matching term

- ☐ Distance the aircraft can fly on the fuel aboard.
- ☐ Engine output at the shaft — cruise fuel burn follows it.
- ☐ Power the engine and propeller can actually deliver.
- ☐ Power needed to hold level flight at a given airspeed.
- ☐ The nose's angle to the horizon.
- ☐ Speed where total drag is lowest; parasite = induced; most efficient.
- ☐ Horizontal distance ÷ vertical distance (e.g. 10:1).
- ☐ Slightly slower than best glide — the most TIME in the air.

A L/D max
B Brake horsepower
C Power required
D Power available
E Glide ratio
F Minimum sink speed
G Range
H Attitude

V-speeds

V-speed	What it means	Where you see it
Vso	_____	Bottom of the _____ arc
Vfe	_____	Top of the _____ arc
Va	_____	Not marked — know it
Vne	_____	The _____ line
Vx	_____	—
Vy	_____	—

Vx vs Vy

	Vx	Vy
Gives the most height per …	_____	_____
Use it to …	_____	_____
Slower or faster airspeed?	_____	_____

Power curve

Power available is …	Result
More than power required	_____
Equal to power required	_____
Less than power required	_____ — rate set by the size of the _____
Throttle at idle	Power available drops to _____; power required stays the _____

Which speed?

You want …	Fly at …
The longest glide distance (reach a field)	_____
The most time in the air	_____
Maximum range on your fuel	_____ — well below normal cruise

Quick check

1. 10:1 glide ratio from 3,000 ft above the ground → about _____ ft of glide. 8:1 from 4,000 ft → _____ ft.
2. 1 kt = 1.15 mph = 1.852 km/h. So 100 kt = _____ mph = _____ km/h.
3. Same power, climb higher → _____, because thinner air = less _____. Why not just carry more fuel for range? Fuel adds _____.

Mock exam questions — circle one answer

1. In straight and level flight with the engine running, the pilot sets the airspeed with: Mock A

a The flaps **b** The engine power / RPM **c** The rudder
d The trim tab

2. The glide ratio of an aircraft is: Mock A

a Height lost divided by distance flown forward
b Distance flown forward divided by height lost
c Airspeed divided by the rate of sink in feet per minute
d Lift divided by weight at the best glide speed

3. After take-off from a long runway with no obstacles, which speed gets the aircraft to cruise altitude in the least time? Mock B

a Vx **b** Vy **c** Vne **d** Vso

4. Total drag is lowest at the airspeed where: Mock B

a The aircraft is about to stall
b Parasite drag is at its maximum
c Parasite drag equals induced drag
d Induced drag drops to zero

5. With the engine failed, the pilot controls airspeed by: Mock B

a Using the rudder **b** Using the throttle
c Retracting the flaps **d** Changing the pitch attitude

6. An aircraft is in level flight at cruise speed. The pilot reduces power and holds the same attitude. The aircraft will: Mock B

a Sink, at a rate set by the power shortfall
b Climb, because the drag on the aircraft is now lower
c Hold its altitude, but at a slower airspeed than before
d Stall at once, regardless of the airspeed it had

7. In cruise, a piston engine's fuel burn is roughly proportional to its: Mock C

a True airspeed **b** Altitude **c** Propeller diameter
d Brake horsepower

8. An aircraft with a 10:1 glide ratio is 3,000 ft above the ground. In still air it can glide about: Mock C

a 300 ft **b** 3,000 ft **c** 13,000 ft **d** 30,000 ft

9. Vx, the best-angle-of-climb speed, gives the most: Mock C

a Altitude gained in a given time
b Altitude gained in a given ground distance
c Airspeed for the power being used
d Range on the fuel carried aboard

10. Match each description to the correct V-speed. Each letter is used once; one letter is not used. Mock A

A Va · **B** Vfe · **C** Vne · **D** Vso · **E** Vx · **F** Vy

- ☐ Never-exceed speed — the red line on the airspeed indicator
- ☐ Maximum speed with the flaps extended — top of the white arc
- ☐ Stall speed in the landing configuration — bottom of the white arc
- ☐ The speed that gains the most altitude per minute
- ☐ The speed the structure is built to take a single full, abrupt control deflection at

Partner challenge (2 min): Engine failure at 3,000 ft, field 5 NM away. Talk your partner through it: which speed do you fly, how do you hold it with no engine, and why not minimum sink?

MODULE 6 Performance: Stall & High-Speed Flight

Full notes, flashcards & mock exams on the site

Big ideas — read, then fold this box back and work from memory

- **Boundary layer:** air speed is zero at the skin and rises to free-stream speed. It gives the wing an **effective shape**.
- **Stall = critical angle of attack exceeded → boundary layer separates.** It's about angle, not speed.
- **Root-first stall is the goal** — the tips and ailerons keep flying. Rectangular wing → root first; tapered/swept → tips first.
- **Vortex generators** stir fast, high-energy air into the boundary layer → it stays attached longer → lower stall speed.
- **Mach = speed ÷ speed of sound.** Colder air = slower speed of sound. Through a shock: **pressure ↑, temp ↑, velocity ↓**.

Vocabulary — write the letter of the matching term

☐ Aircraft speed ÷ the speed of sound.
☐ Where on the wing the stall starts first.
☐ Slanted shock attached to a sharp leading edge; flow stays supersonic.
☐ Speed a small pressure wave travels; ~661 kt at standard sea level.
☐ A shock wave separates the boundary layer — a stall caused by speed.
☐ About Mach 0.8 to 1.3 — mixed subsonic and supersonic flow.
☐ Boundary layer lifting off the surface — the cause of a stall.
☐ Angled-back wing that lowers the effective Mach number along the chord.

A Separation
B Stall pattern
C Mach number
D Speed of sound
E Transonic
F Shock stall
G Oblique shock
H Sweepback

Through a shock wave — up or down?

Pressure	Temperature	Velocity	Density
_____	_____	_____	_____

Two kinds of stall

	Normal stall	Shock stall
Caused by	_____	_____
Angle of attack	_____	_____
Speed	_____	_____

Stall patterns

Wing shape	Stalls first at the …
Rectangular (straight)	_____
Tapered or swept	_____
Desirable pattern	_____ — because the _____ keep working and the pilot gets a _____ warning

High-speed flight

Term	Fill it in
Critical Mach (Mcrit)	Aircraft speed when air over the wing first reaches _____
Transonic range	About Mach _____ to Mach _____
Wave drag	Big drag rise caused by _____ in the transonic range
Sweepback	_____ critical Mach — less airflow crosses the _____
Thin wing, sharp leading edge	Shock attaches as an _____ shock → far less wave drag

Quick check

1. Speed of sound 661 kt, aircraft at 330 kt → Mach _____. Higher up the air is colder, so the speed of sound is _____.

2. 14 Oct 1947: Capt. _____ flew the Bell _____ past Mach 1.

Mock exam questions — circle one answer

1. The boundary layer is the thin layer of air next to a surface in which: Mock A

a Static pressure rises from zero up to the free-stream pressure value
b Air speed rises from zero at the skin to the free-stream speed
c Temperature falls steadily from the hot surface to the free stream
d All of the lift is produced by the camber of the upper surface

2. As air passes through a shock wave, its velocity: Mock A

a Increases, and pressure and temperature increase
b Increases, and pressure and temperature decrease
c Decreases, and pressure and temperature increase
d Stays the same while pressure and temperature increase

3. Designers want a wing to stall at the root first because: Mock A

a The root is the strongest part of the wing structure
b The root is where the fuel tanks and the landing gear are
c The tips and ailerons keep flying, so roll control is kept
d It gives the passengers the smoothest possible ride

4. Vortex generators delay the stall by: Mock B

a Mixing high-energy air into the boundary layer
b Heating the leading edge to stop ice forming
c Adding induced drag on purpose to slow the wing
d Reducing the wing area that meets the airflow

5. The speed of sound in the atmosphere depends mainly on the air's: Mock B

a Pressure **b** Oxygen content **c** Temperature
d Humidity

6. A wing stalls when: Mock B

a The airspeed falls below the bottom of the white arc on the indicator
b The engine stops producing power while the aircraft is in a climb
c The pilot extends full flap at a speed that is far above Vfe
d The critical angle of attack is exceeded and the airflow separates

7. Sweeping a wing back raises its critical Mach number because: Mock B

a A swept wing is thinner and lighter than a straight one of the same span
b The wing area is reduced, which lowers the total drag on the aircraft
c The leading edge becomes sharper, so a bow wave cannot form ahead of it
d Less of the airflow crosses the chord, so shocks form later

8. A rectangular (straight, untapered) wing tends to stall first at the: Mock C

a Tip **b** Root **c** Leading edge slat **d** Aileron

9. The transonic speed range is roughly: Mock C

a Mach 0 to 0.5 **b** Mach 0.8 to 1.3 **c** Mach 1.3 to 5
d Mach 2 to 3

10. Match each description to the correct high-speed flight or stall term. Each letter is used once; two letters are not used. Mock C

A Boundary layer · **B** Critical Mach number · **C** Shock wave · **D** Vortex generator · **E** Wave drag · **F** Sweepback · **G** Shock stall

- ☐ Thin layer next to the skin where the air speed rises from zero to the free-stream value
- ☐ The aircraft speed at which the air flowing over the wing first reaches Mach 1
- ☐ Very thin wall in the flow where velocity drops and pressure and temperature jump
- ☐ Small fin that stirs high-energy air into the boundary layer to delay separation
- ☐ The sharp rise in drag caused by shock waves in the transonic range

Partner challenge (2 min): Draw a wing on scrap paper. Partner A shows a normal stall, Partner B shows a shock stall. Each explains what the boundary layer is doing.